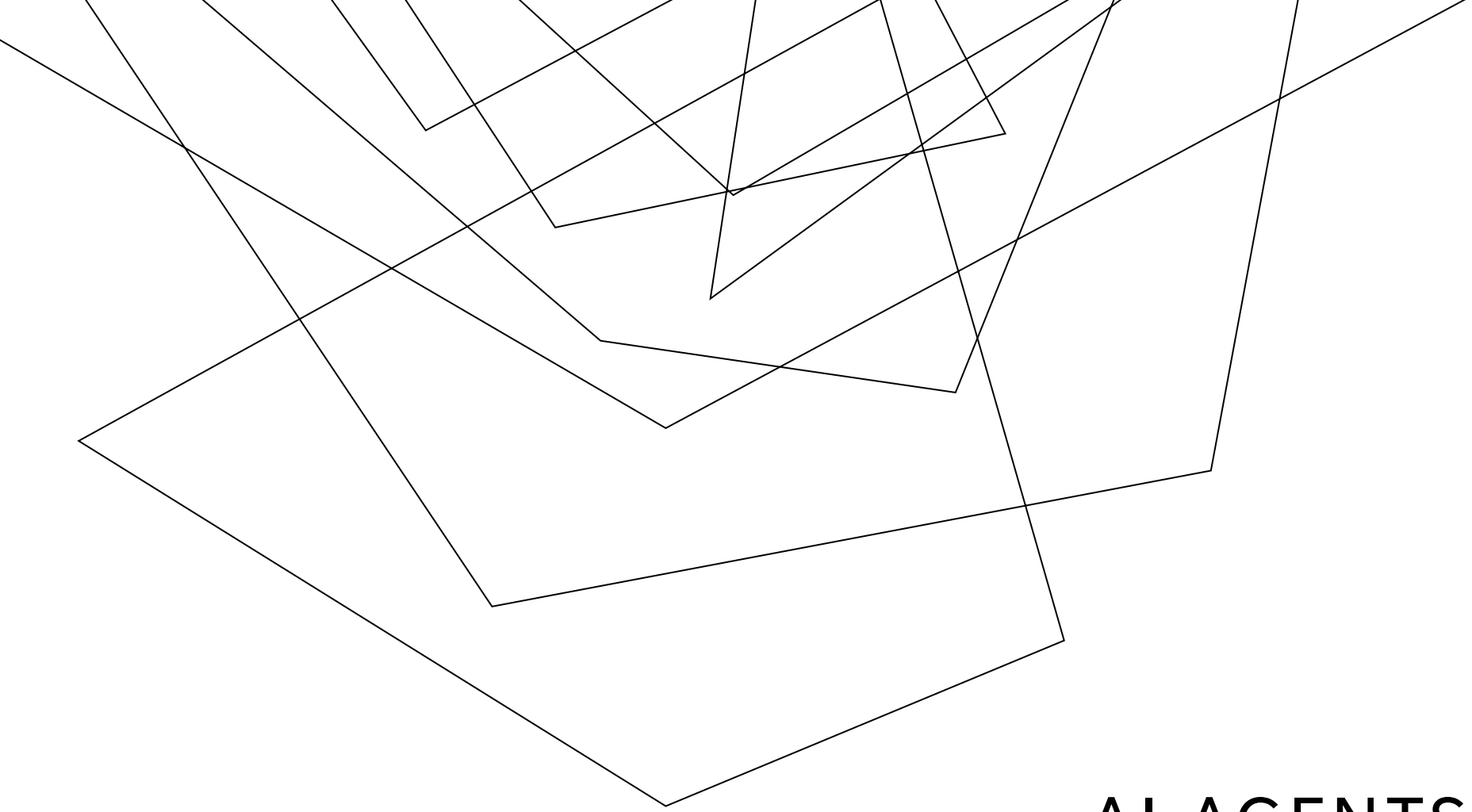


AI AGENTS

What *is* an AI agent?

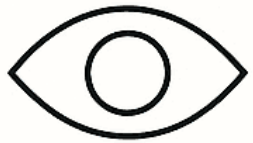
A program that **autonomously** completes tasks or makes decisions based on data.

AI Agents- The multi-trillion dollar opportunity

Picture this: your assistant knows to open an umbrella when it rains. But what if the weather changes, and it gets extremely sunny? How would they behave in this scenario depending on whether their role is that of a bot, a copilot, or an AI agent?

Aspect	Bots	Copilots	AI Agents
	☂️🌧️ | 🌂☀️	🌂☀️❓	☂️🌧️ | ☂️☀️
Behaviour in given scenario	Opens the umbrella only when it rains, as that's the defined scope.	Suggests using the umbrella for sun protection but waits for the user to approve.	Recognizes the heat, evaluates the umbrella's material, and autonomously opens it for shade.
Definition	Programs designed to handle predefined queries and responses based on specific scopes.	Tools designed to assist humans in completing tasks, often providing contextual suggestions.	Autonomous systems capable of understanding, adapting to new situations, and executing tasks.
Scope of Knowledge	Limited to a fixed scope defined by pre-set rules or training data.	Broader than chatbots, but still reliant on direct prompts and user guidance.	Broad and dynamic; capable of integrating contextual data and external resources.
Adaptability	Cannot adapt to situations outside the pre-defined scope.	Provides intelligent suggestions but relies on human intervention for decision-making.	Fully autonomous; adapts to new circumstances using reasoning and contextual knowledge.
Decision making	No decision-making capabilities; strictly follows pre-programmed logic.	Suggests decisions but depends on users to make final choices.	Independent decision-making based on analysis of context and goals.
Execution of tasks	Executes simple, predefined tasks upon receiving specific inputs.	Assists users in executing tasks but does not act autonomously.	Plans and executes complex tasks autonomously with minimal input.
Learning capability	Static; requires reprogramming to learn or expand capabilities.	Dynamic; learns from user inputs and improves suggestions over time.	Dynamic and self-sufficient; integrates external knowledge and evolves with changing contexts.

THE CORE COMPONENTS OF AI AGENTS

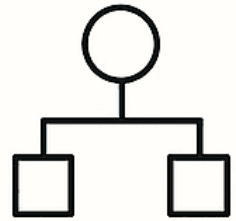


PERCEPTION

How AI takes in the world

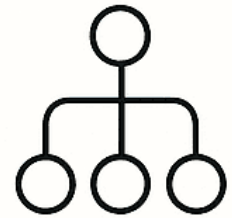
KNOWLEDGE BASE & MEMORY

The brain behind the bot



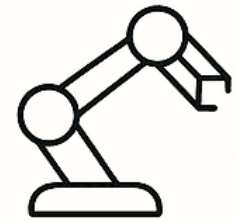
REASONING & DECISION-MAKING

The engine that chooses what to do next



LEARNING MECHANISM

Where code starts to feel smart



ACTION & EXECUTION

Turning thought into resuits

What *is* an AI agent?

NON-AGENTIC

(One-shot)

Write an article on topic from start to finish

AGENTIC

Write an outline on topic

Does this need online research?

Write a first draft article on topic

Evaluate what needs more work

Revise the draft further

AGENTIC CODING

Reflection, planning, multi-agent...

Write code that does X

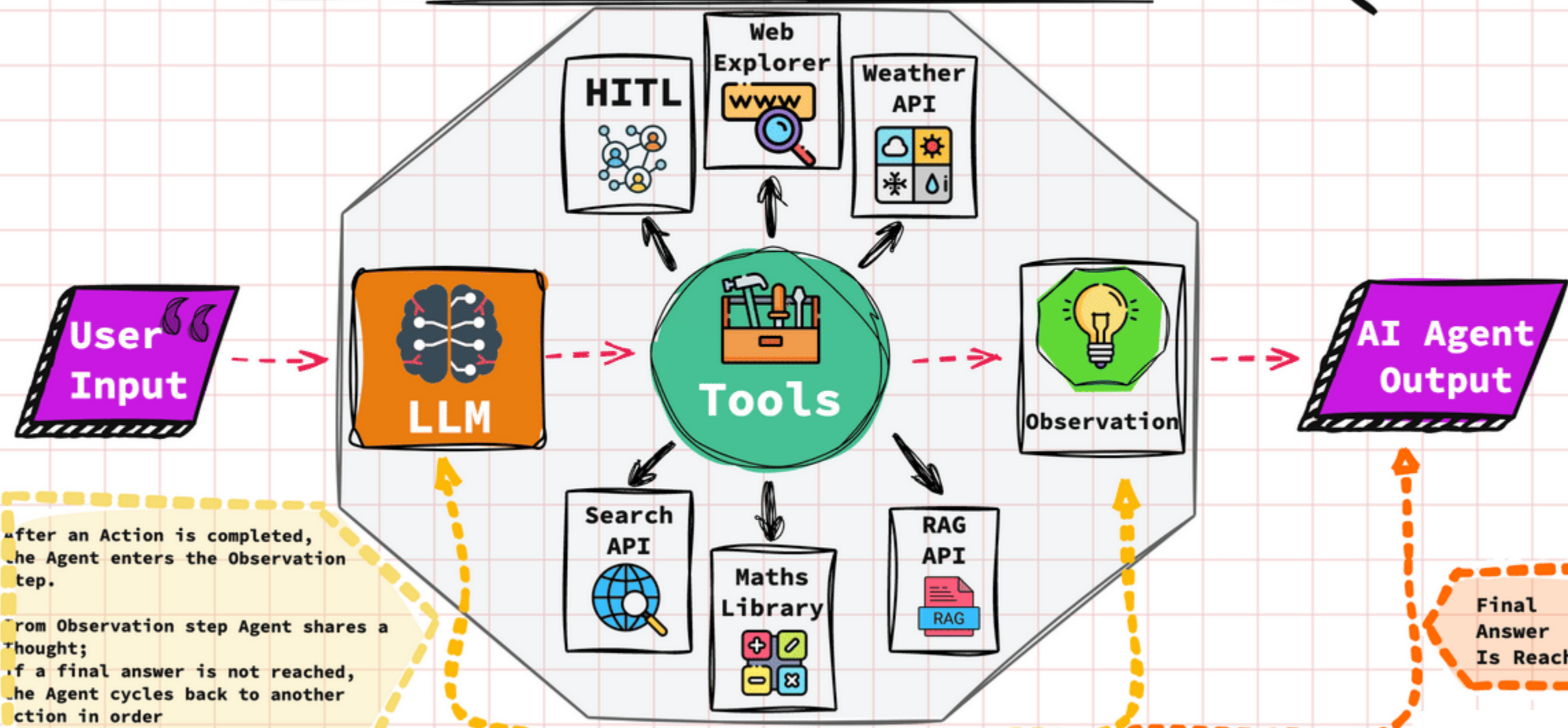
Version 1 Line 8 has a bug. Fixing...

Version 2 Test Y failed. Fixing...

Version 3 All issues should be fixed ✅

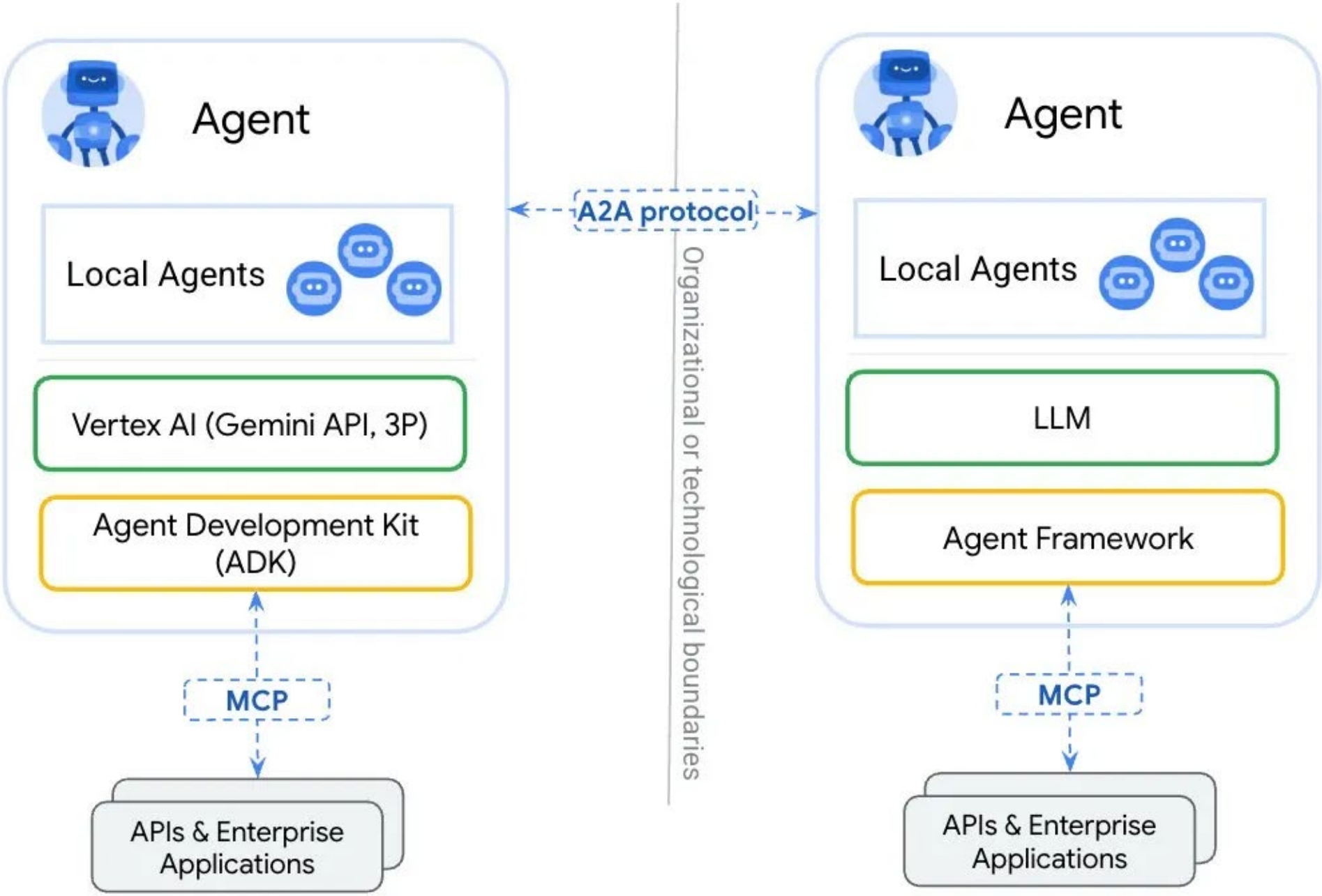
OK

AI AGENTS - AGENTIC APPLICATIONS



fter an Action is completed,
he Agent enters the Observation
tep.

rom Observation step Agent shares a
hought;
f a final answer is not reached,
he Agent cycles back to another
ction in order
o move closer to a Final Answer.


Agent
Local Agents
Vertex AI (Gemini API, 3P)
Agent Development Kit (ADK)
MCP
APIs & Enterprise Applications
A2A protocol
Organizational or technological boundaries
Agent
Local Agents
LLM
Agent Framework
MCP
APIs & Enterprise Applications